Integrative Multi-omics Summary-based Mendelian Randomization Identifies Key Oxidative Stress-related Genes As Therapeutic Targets For Atrial Fibrillation And Flutter

Reviewed By Melika Janmohammadi Firoz

B.Sc. in Cellular and Molecular Biology, Azad University, Qaemshahr branch

Background

- ✓ Atrial fibrillation (AF) is a prevalent cardiac arrhythmia associated with substantial morbidity and mortality.
- ✓ Oxidative stress (OS) has been implicated in the **pathogenesis of AF**, suggesting that targeting OS-related genes could offer novel therapeutic opportunities.
- ✓ This study aimed to **identify causal OS-related genes contributing to AF** through a comprehensive multi omics Summary-based Mendelian Randomization (SMR) approach.

Methods:

- This study **integrated data from genome-wide association studies (GWAS)** with methylation quantitative trait loci (mQTL), expression QTL (eQTL), and protein QTL (pQTL) to explore the relationships between oxidative stress-related (OS-related) genes and AF risk.
- Genes associated with oxidative stress and AF were obtained from the **Nielsen et al. study (discovery) and the FinnGen study (replication).**
- The **SMR analysis and HEIDI test** were utilized to assess causal associations, followed by **Bayesian co-localization analysis (PPH4 > 0.5)** to confirm shared causal variants.
- **Multi-omics data were employed to analyze the associations within mQTL-eQTL** pathways.
- **A two-sample MR analysis was conducted for sensitivity** verification.
- The significance of findings was determined using a false discovery rate **(FDR) < 0.05 and p_HEIDI > 0.01.**

Results:

- **At the DNA methylation level, 19 CpG sites near 7 unique genes** were found to have causal effects on AF and strong co-localization evidence support ($PPH_4 > 0.70$).
- At the gene expression level, **six oxidative stress-related genes from eQTL Gen and three from GTEx (v8)**, including TNFSF10, CDKN1A, ALOX15, TTN, PTK2, ALB, KCNJ5, and CASQ2, were found to have causal effects on AF in the sensitivity and co-localization analyses ($PPH_4 > 0.50$).
- At the circulating protein level, both ALAD (OR 0.898, 95% CI 0.845–0.954, $PPH_4 = 0.67$) and APOH (OR 0.896, 95% CI 0.844–0.952, $PPH_4 = 0.93$) were associated with a lower risk of AF, and APOH was validated in the replication group.
- After **integrating the multi-omics data between mQTL and eQTL,** we identified two oxidative stress-related genes, **TTN** and **CASQ2**. The methylation of cg09915519 and cg10087519 in TTN was associated with higher expression of TTN and a lower risk of AF, which aligns with the negative effect of TTN gene expressionon AF risk. TTN may play a protective role in AF.

Conclusion:

- This study identified several OS-related genes, particularly TTN, as having causal roles in AF, which were verified across three-omics pathways.
- The findings underscore the importance of these **genes in AF pathogenesis and highlight their potential as therapeutic targets.**
- The integration of multi-omics data provides a comprehensive **understanding of the molecular mechanisms underlying AF,** paving the way for targeted therapeutic strategies.

What is the Problem?

- The pathogenesis of AF **is complex**, involving various factors such as electrical and mechanical changes in the atria, inflammation, oxidative stress, and fibrosis (Gaudino et al., 2023).
- Inflammation has been well-documented as a critical factor in the development and progression of AF, leading to atrial fibrosis and structural remodeling.
- Similarly, OS, characterized by the excessive production of reactive oxygen species (ROS), has been implicated in atrial remodeling and the onset of AF (Ramos-Mondragón et al., 2023).
- Previous studies have highlighted **the interplay between inflammation and OS in the** pathophysiology of AF (Ramos Mondragón et al., 2023; Balan et al., 2024).
- Inflammatory markers such as high-sensitivity C-reactive protein and galectin-3 have been associated with AF incidence and recurrence, suggesting their role in atrial fibrosis and remodeling (Noubiap et al., 2021).
- Additionally, OS markers like myeloperoxidase and oxidized low-density lipoprotein have been linked to cardiac tissue damage and AF (Rafaqat et al., 2021).
- Despite these associations, the relative **contributions of inflammation and OS to AF and their potential interactions remain not fully understood.**

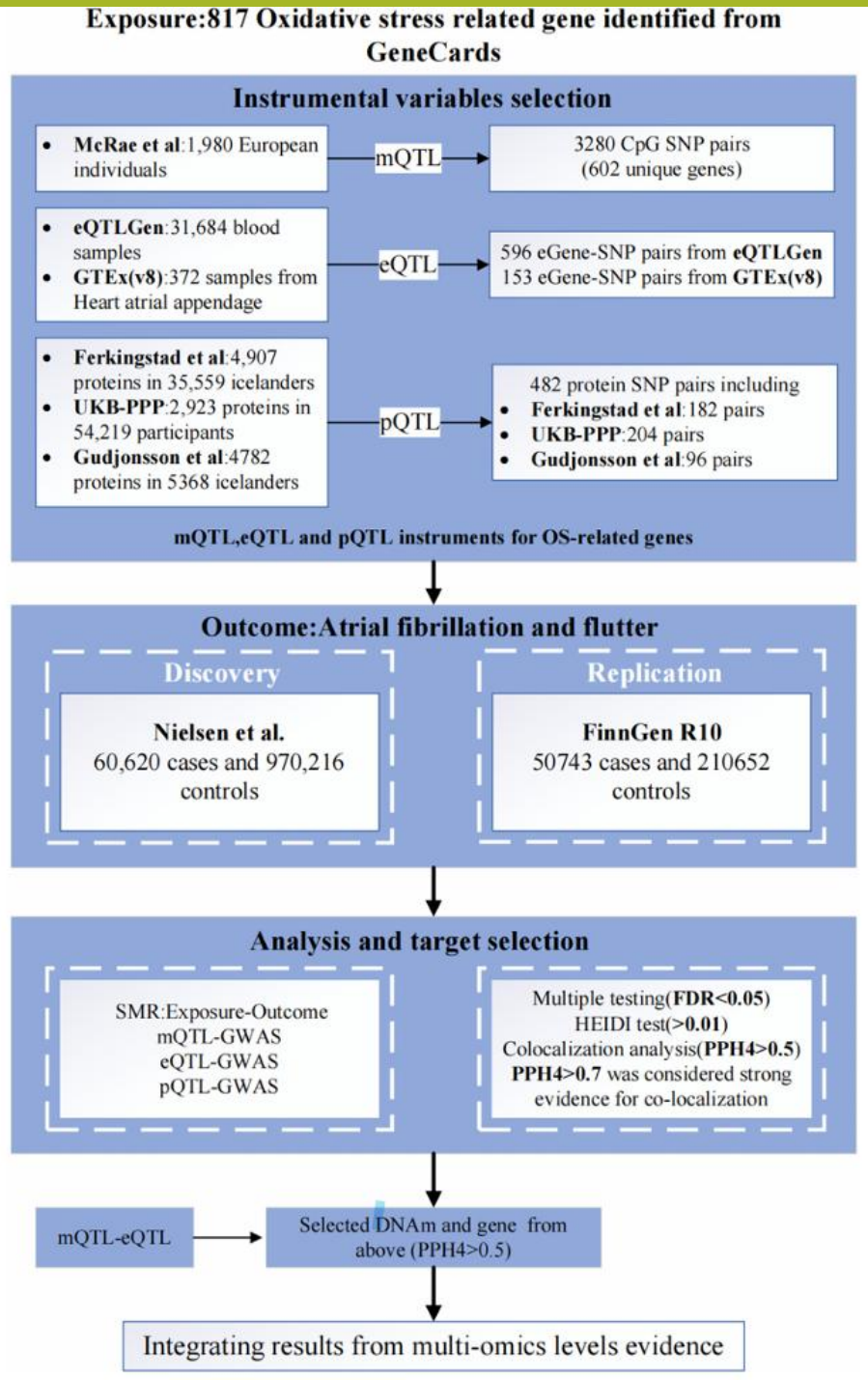
Solution and hypothesis of the study

- Recent advancements in multi-omics technologies provide a unique opportunity to investigate the **molecular mechanisms underlying AF** by integrating genomic, epigenomic, and proteomic data.
- **Summary-based Mendelian randomization is a powerful analytical approach** that leverages genetic variants as instrumental variables to infer **causal relationships between risk factors and disease outcomes.**

➢ **We hypothesized that OS related genes play a causal role in AF and that their identification could provide novel targets for therapeutic intervention.**

- **By combining SMR with multi omics data,** it is **possible to identify causal genes and pathways involved in AF**, therapeutic targets. offering insights into potential In this study, we aimed to identify therapeutic genes for AF by GWAS with mQTL, eQTL, and pQTL data using a comprehensive SMR approach.

- Firstly, they identified 817 OS-related genes from the GeneCards database.
- Next, we integrated GWAS of AF with blood expression QTL (eQTL), methylation QTL (mQTL), and protein QTL (pQTL) summary data for analysis using the SMR method.
- Then, we identified potential causal OS-related genes through **SMR analysis and Bayesian co-localization analysis**
- we also analyzed the molecular networks of OS-related genes within mQTL and eQTL through SMR analysis.
- Finally, we validated the robustness of the primary findings using.



Data source

GWAS data:

- A two-sample MR analysis, The GWAS by **Nielsenetal**.encompassed1,030,836participants (60,620 AF cases and 970,216 controls) of **European descent** (Nielsen et al., 2018). The median age was not provided, and 53% of the participants were women.
- Summary-level data of genetic associations with AF were also obtained from the publicly available data release of the **FinnGen study** (Kurki et al., 2023).

The diagnosis of AF was based on ICD codes and confirmed by Social Insurance Institution codes, with a **total of 50,743 AF cases and210,652 controls.**

QTL data:

McRaeetal. reported data on **SNP-CpGassociations** in blood from a study of 1,980 individuals of European descent, focusing on mQTLs (McRae et al., 2018)).

Gene expression data were sourced from the eQTLGen consortium, which offered a substantial sample size (n = 31,684) To identify SNPs associated with the expression of genes targeted by the **corresponding plasma proteins** (Võsa et al., 2021).

For protein abundance, summary statistics were obtained **from three independent pQTLstudies** conducted by **Ferkingstadet al., Gudjonsson et al., and the UKB-PPP.**

The **tissue-specific expression** of target genes that could potentially cause AF was further assessed using tissue-specific QTL data from the Genotype Tissue Expression **(GTEx) web portal**.

extensive resource enabled the exploration of **gene expression across a variety of tissues,** providing insights into the role of specific genes in AF pathogenesis.

OS-related genes data:

- OS-related genes were identified from the **GeneCards database** (version 5.10, https://www.genecards.org) using the keyword "oxidative stress" .
- 817 OS-related genes. After the **filtering process** for OS-related genes, 602 genes is identified and associated with methylation, 596 with expression from eQTLGen, 153 with expression from GTEx (v8), and 482 proteins with available instruments (mQTLs, eQTLs, and pQTLs withP

Summary-data-based MR analysis

- SMR analysis was employed to assess the association between OS-related gene methylation, expression, and protein abundance with the risk of atrial fibrillation and flutter (Zhu et al., 2016).
- We constructed a hypothetical model of the mediation mechanism in which a single SNP influences a trait by altering the DNA methylation (DNAm) level, which in turn regulates the expression levels of a functional gene. Therefore, we also used the SMR to analyze the OS-related DNAm and gene within the multi-omics respectively (mQTL-eQTL).
- **Utilizing the top associated cis-QTLs, the SMR method achieved significantly greater statistical power compared to conventional MR analysis, particularly when exposure and outcome data are derived from two independent samples with large sample sizes. The top associated cis QTLs were selected by considering a window centered around each relevant gene (±1,000 kb) and meeting a p-value threshold of 5.0×10^{-8}. SNPs with allele frequency differences exceeding the specified threshold (0.2 in this study).**
- **The heterogeneity in the dependent instrument (HEIDI) test was employed to distinguish between pleiotropy and linkage, with P_HEIDI < 0.01 indicating likely pleiotropy, leading to exclusion from the analysis.**
- **P-values were adjusted to control the false discovery rate (FDR) at $\alpha = 0.05$ using the Benjamini–Hochberg method. The SMRanalysis and HEIDI test were performed using version 1.3.1 of the SMR software.**
- **Two-sample MR analysis were conducted using the "TwoSampleMR (version 0.5.6)" package of the R software (version 4.2.2).**

Co-localization analysis

Five distinct posterior probabilities are reported, corresponding to the following five hypotheses:

- 1) no causal variants for either of the two traits (H0)
- 2) a causal variant for gene expression only (H1)
- 3) a causal variant for disease risk only (H2)
- 4) distinct causal variants for each trait (H3)
- 5) a shared causal variant for both traits (H4).

Integrating results from multi-omics level of evidence

To obtain a full picture of the associations between the **regulation of OS-related genes and AF at the genomic level,** we conducted SMR analysis of the causal associations within OS-related gene methylation and expression, in order to explore the basic mechanism **(FDR < 0.05, P_HEIDI > 0.05).**

- **We focused only on the genes that passed the screening criteria. The identification of the final putative causal relationships was defined as:**
- 1) false discovery rate (FDR) < 0.05 in all three-step SMR
- 2) P_HEIDI > 0.01 in mQTL-GWAS and eQTL GWAS SMR,P_HEIDI>0.05 in mQTL-eQTLSMR
- 3) PPH4 >0.5 in co-localization analysis of eQTL and AF GWAS
- 4) the eQTL and mQTL should correspond to the same gene symbol.

Results

Oxidative stress-related gene methylation and AF GWAS data

Results for causal effects of oxidative stress-related gene methylation on AF are visualized in (Table 1).

The direction of effect estimates were not always consistent for different CpG sites located in the same gene. For example, one SD increase in geneticall predicted **MAPT** methylation atcg02228913 was associated with a **decreased risk of AF**(OR0.967, 95%confidence interval [CI] 0.955–0.979), whereas one SD increase ingenetically predicted **MAPT** methylation at cg07163735 was associated with a **higher risk of AF** (OR1.189, 95%CI 1.101–1.284).

- **Among these identified CpG sites, the associations for cg09915519 and cg10087519 near TTN were replicated in Finn Gen with strong colocalization**

TABLE 1 Oxidative stress gene methylation and AF GWAS data.

Gene	ProbeID	TopSNP	β coefficient	Or (95% CI)	FDR	*p* HEIDI	PPH4
MAPT	cg02228913	rs112572874	−0.033	0.967 (0.955.0.979)	2.27E-05	0.01	0.77
	cg21705961	rs17689,882	−0.063	0.939 (0.917.0.961)	3.52E-05	0.10	0.82
	cg05301556	rs10445363	−0.081	0.923 (0.894.0.952)	1.35E-04	0.14	0.83
	cg23202277	rs112995313	0.117	1.124 (1.073.1.178)	1.59E-04	0.23	0.83
	cg05772917	rs112155389	0.156	1.169 (1.092.1.251)	9.97E-04	0.04	0.83
	cg07163735	rs538628	0.173	1.189 (1.101.1.284)	1.29E-03	0.10	0.80
	cg01934064	rs2668668	0.089	1.093 (1.042.1.145)	1.67E-02	0.14	0.79
	cg00846647	rs111751251	−0.038	0.963 (0.943.0.983)	2.16E-02	0.09	0.76
CRHR1	cg16228356	rs17689,882	−0.051	0.951 (0.933.0.968)	2.92E-05	0.05	0.83
	cg23762722	rs112746008	−0.078	0.925 (0.898.0.953)	4.34E-05	0.09	0.78
	cg05727186	rs62056790	−0.095	0.91 (0.877.0.944)	1.14E-04	0.12	0.81
	cg24063856	rs62057073	−0.140	0.87 (0.817.0.926)	1.63E-03	0.08	0.73
	cg16642545	rs111273167	−0.172	0.842 (0.769.0.923)	1.67E-02	0.03	0.77
TTN	cg09915519	rs744426	−0.256	0.774 (0.713.0.84)	6.07E-07	0.27	0.93
	cg10087519	rs12998857	−0.364	0.695 (0.597.0.808)	3.65E-04	0.29	0.93
ALPP	cg14659346	rs1048995	−0.037	0.964 (0.945.0.984)	2.29E-02	0.61	0.80
CREBBP	cg05194552	rs2530890	−0.149	0.862 (0.794.0.935)	2.29E-02	0.26	0.96
GPX4	cg04903600	rs62131210	0.095	1.099 (1.045.1.157)	1.67E-02	0.05	0.83
MAP2K2	cg21124940	rs60505738	−0.037	0.963 (0.944.0.983)	1.68E-02	0.95	0.85

SNP, single nucleotide polymorphisms; OR, odds ratio; CI, confidence interval; FDR, false discovery rate; PPH4, posterior probability of H4.

Oxidative stress-related gene expression and AF GWAS data

Results for causal effects of oxidative stress related gene expression on AF are presented in (Table 2).

TABLE 2 Oxidative stress gene expression and AF GWAS data.

Gene	TopSNP	β coefficient	Or (95% CI)	FDR	*p* HEIDI	PPH4
TTN	rs13034990	−1.235	0.291 (0.186.0.454)	1.48E-05	0.04	0.65
PTK2	rs10088133	−0.230	0.794 (0.73.0.864)	1.48E-05	0.41	0.67
CDKN1A	rs12199346	0.194	1.214 (1.125.1.309)	5.17E-05	0.02	0.64
ALB	rs150643605	−0.421	0.656 (0.529.0.815)	7.89E-03	0.04	0.98
TNFSF10	rs3181140	0.149	1.16 (1.074.1.254)	9.64E-03	0.59	0.65
ALOX15	rs72835630	0.061	1.063 (1.028.1.099)	1.54E-02	0.47	0.60

SNP, single nucleotide polymorphisms; OR, odds ratio; CI, confidence interval; FDR, false discovery rate; PPH4, posterior probability of H4.

In total, 83 associations were identified. After multiple testing correction and co-localization analysis, we selected those genes which PPH4 > 0.5.

Genetically predicted higher levels expression of **TNFSF10** (OR 1.160, 95% CI 1.074–1.254; PPH4 = 0.65), **CDKN1A** (OR 1.214, 95% CI 1.125–1.309; PPH4 = 0.64) and **ALOX15**(OR 1.063, 95% CI 1.028–1.099; PPH4 = 0.60) **were positively associated with AF risk.**

Conversely, genetically predicted higher levels expression of **TTN** (OR 0.291, 95% CI 0.186–0.454; PPH4 = 0.65), PTK2 (OR 0.794, 95% CI 0.730–0.864; PPH4 = 0.67) and **ALB**(OR 0.291, 95% CI 0.186–0.454; PPH4 = 0.98) **were inversely associated with AF risk** (Table 2).

Associations for TTN and PTK2 were all replicated in Finn Gen, of which the genetically predicted high level expressions **were negatively associated with AF risk** (OR 0.181, 95% CI 0.098–0.336; OR 0.744, 95% CI 0.661–0.837, respectively). **whereas CDKN1A replicated in Finn Gen was positively associated with AF risk** (OR1.36,95%CI1.21–1.529, SupplementaryTableS7).

Tissue-specific validation

Causal associations of the expression of identified genes with AF were further explored in atrial tissue.

TABLE 3 Tissue-specific oxidative stress-related gene expression and AF GWAS data.

Gene	Tissue	TopSNP	β coefficient	Or (95% CI)	FDR	*p* HEIDI	PPH4
KCNJ5	Atrial appendage	rs78907918	0.257	1.293 (1.169.1.431)	9.85E-05	0.06	0.97
CASQ2	Atrial appendage	rs4073778	−0.428	0.652 (0.538.0.789)	7.18E-04	0.64	0.98
ALOX15	Atrial appendage	rs8071990	−0.089	0.915 (0.867.0.966)	3.62E-02	0.54	0.76

SNP, single nucleotide polymorphisms; OR, odds ratio; CI, confidence interval; FDR, false discovery rate; PPH4, posterior probability of H4.

In total, 22 associations were identified to be associated with AF at the nominally significant level (PSMR < 0.05, P HEIDI > 0.01).After multiple testing correction and co-localization analysis (Supplementary Table S9), genetically predicted expression levels of **KCNJ5** was associated with **increases in AF risk in atrial appendage** (OR 1.293, 95% CI 1.169–1.431; PPH4 = 0.97). In contrast, genetically predicted expression levels of **CASQ2 and ALOX15 were associated with reduction in AF risk inatrial appendage tissue** (OR0.652, 95%CI 0.538–0.789; PPH4 = 0.98 and OR 0.915, 95% CI 0.867–0.966; PPH4 = 0.76, respectively). **KCNJ5 was replicated in the FinnGen study.**

Oxidative stress-related gene relative proteins and AF GWAS data

- Oxidative stress-related gene relative proteins and AF GWAS data There **were 36 oxidative stress related proteins** from 3 independent researches i**ncluding Ferkingstad study, Gudjonsson study and UKB-PPP Olink** which separately associated with AF risk at PSMR < 0.05 level.
- After adjustment for multiple testing, we obtained 2 proteins including ALAD, APOH (Supplementary Table S12). **Both ALAD (Ferkingstad study) and APOH(Gudjonsson study) were associated with a lower risk of AF**(OR 0.898, 95% CI 0.845–0.954, PPH_4 = 0.67; OR 0.896, 95% CI 0.844–0.952, PPH_4 = 0.93, respectively). **APOH was replicated in the FinnGen study**

SMR analysis for mQTL and eQTL dat

- **It is well known that gene methylation affects gene expression.** Therefore, we proceeded to explore the **possible link between DNAm and gene expression by using DNAm as the exposure and transcripts as the outcome.**
- After screening the results by FDR < 0.05 and P HEID > 0.05, we obtained the regulatory relationships for the expression of **252 OS-related genes regulated by 595 DNA methylation CpGsites** (Supplementary Table S14). We only analyse the genes which have higher co-localization evidence (PPH4 >0.5) support with AF.

Finally, we identified **1 OS related genes regulated by 2 DNA methylation CpG sites (Figure 2).** TTN, there were two significantly associated methylation sites (cg09915519 and cg10087519), **two of which were positively correlated with TTN expression and passed the co-localization analysis in mQTL GWAS (Figure 3).**

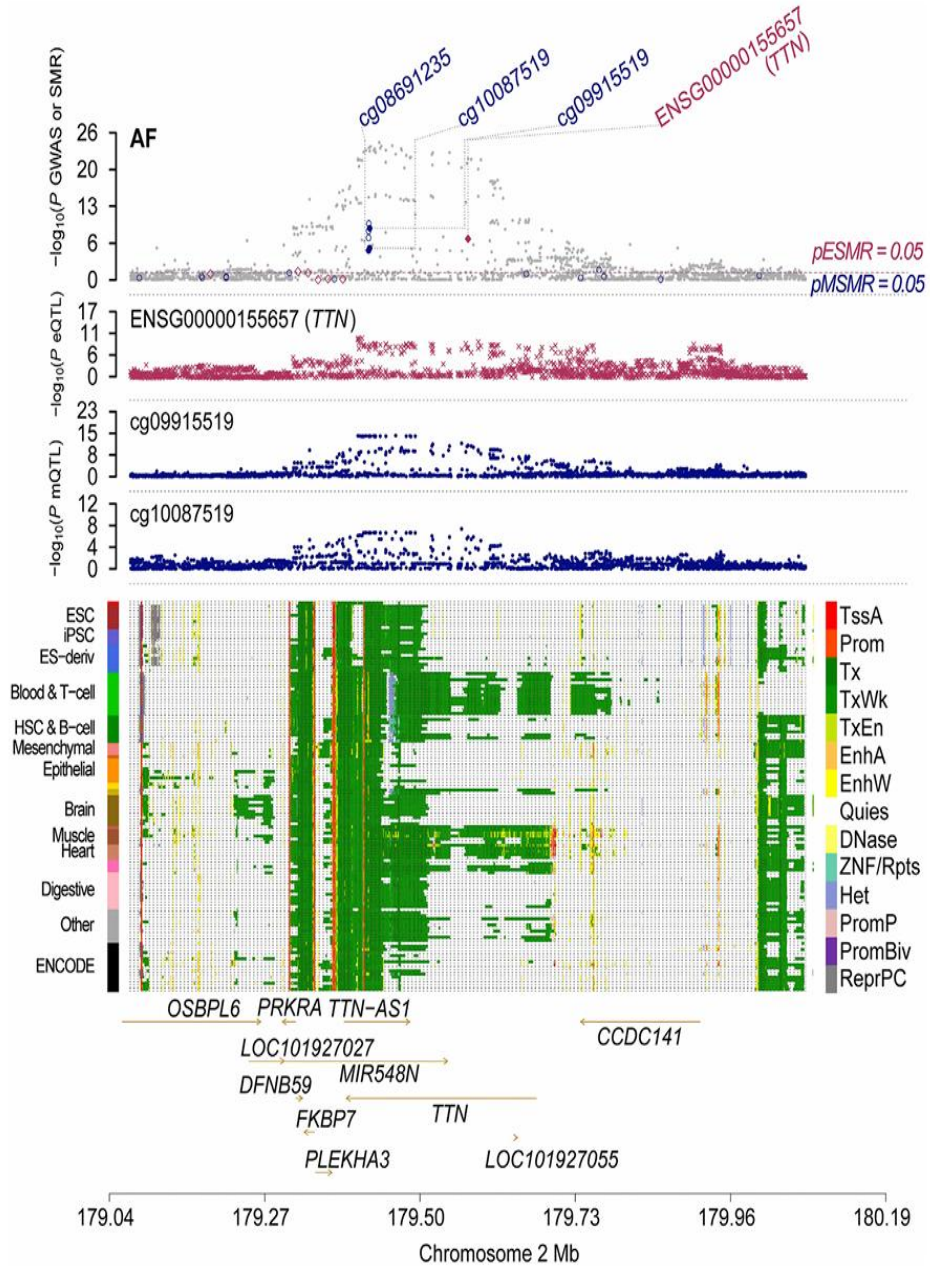


FIGURE 2
Multi-omics data integration for AF. Results of associations across mQTL, eQTL and GWAS at the TTN locus.

SMR analysis for mQTL and eQTL from GTEx(v8) tissue data

After screening the results by **FDR < 0.05 and P HEID >0.05**, we obtained the regulatory relationships for the expression of **92 OS related genes from tissue of heart atrial appendage regulated by 256 DNA methylation CpG sites** (Supplementary Table S13).

We analyse the genes which have co-localization evidence (PPH4>0.5) support with AF.

Finally we identified CASQ2 negatively regulated by cg20810993 CpG sites.

Multi-omics level integration

After integrating evidence from multi-omics data, we identified **TTN from eQTLGen and CASQ2 from GTEx (v8) with higher evidence linking them to AF through the mQTL to eQTL pathway**.

We constructed a hypothetical model of the mediation mechanism: **a single nucleotide polymorphism (SNP) affects thetrait by altering DNA methylation (DNAm) levels,** which then **regulate the expression of a functional gene**.

One CpG sites (cg20810993) were excluded because it did not show significant results in them QTL-GWAS SMR analysis.

In conclusion, the integration analysis results demonstrated **that TTN methylation regulated by cg09915519 and cg10087519 increasesTTN expression,which is associated witha lower risk of AF(Figure4).**

Discussion

- TTN (Titin) is a major **structural protein in cardiac muscle and is essential for normal heart contraction**.
- TTN truncating variants (TTNtv) are strongly associated with:
  - cardiomyopathy
  - arrhythmias
  - heart failure
  - early-onset Atrial Fibrillation
- TTN mutations can cause:
  - sarcomere dysfunction
  - atrial dilation and fibrosis
  - impaired protein degradation pathways
  - increased cellular stress
- Previous studies identified TTN as one of the **most frequently mutated genes in AF and cardiomyopathy patients**.
- **DNA methylation regulates TTN expression, especially at**:
  - cg09915519
  - cg10087519
- Increased methylation at these CpG sites was associated with:
  - higher TTN expression
  - lower AF risk
- Multi-omics SMR results suggest a **protective role of TTN expression against AF.**
- **Findings support the hypothesis that some TTN-related regulatory variants may reduce AF susceptibility.**

Future therapeutic potential:

- epigenetic targeting
- gene editing approaches such as CRISPR-Cas9.

CASQ2 Findings

• CASQ2 (Calsequestrin 2) expression was significantly associated with lower risk of Atrial Fibrillation OR = 0.652, PPH_4 = 0.98

• CASQ2 regulates calcium storage and release in cardiac muscle cells and is essential for calcium homeostasis.

• **Abnormal calcium handling can increase susceptibility to cardiac arrhythmias and AF.**

• Altered CASQ2 expression may **destabilize intracellular calcium signaling and promote abnormal atrial electrical activity.**

• Expected methylation site (cg20810993) regulating CASQ2 **was not validated in discovery/replication analyses**.

APOH Findings

APOH (Apolipoprotein H) was associated with reduced AF risk

oPPH4 = 0.93 replicated in FinnGen cohort

•APOH is involved in:

•Previous studies reported in lipid metabolism

immune regulation

olipoprotein [Lp(a)] pathways

consistent associations between Lp(a) levels and AF risk.

•The APOH–AF causal relationship remains controversial and requires larger validation studies.

Limitation of study

- Most datasets included individuals of European ancestry → limited generalizability.
- Limited availability of OS-related proteins in pQTL datasets.
- Blood-derived genetic data may not fully represent myocardial tissue biology.
- Analysis included only cis-eQTL and cis-mQTL data.
- Cross-sectional design limits evaluation of AF progression over time.
- Experimental/functional validation is still required.

Conclusion

- Multi-omics integration with GWAS provides a powerful framework to identify potential causal genes and therapeutic targets in AF.

Thank you for your attention